SciFig-Eval: A Multilingual Benchmark and Behavioural Framework for Evaluating Vision-Language Models on Scientific Figures

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Abstract

Vision-language models are rapidly becoming integral to the scientific research workflow, supporting tasks from automated literature review and paper evaluation to research assistance and data interpretation. Central to these capabilities is the ability to comprehend scientific figures, which typically encode the core quantitative arguments of published research. As adoption accelerates, it becomes important to investigate how well these models comprehend such visuals, and beyond simple task accuracy, to distill their behavioural profiles (encompassing epistemic honesty, adversarial robustness, and contextual reasoning) that current benchmarks leave largely unexamined.

To address this gap, we introduce SciFig-Eval, a multilingual evaluation corpus of 1,005 scientific figures drawn from over 300 research papers across four languages (English, Bulgarian, Chinese, German) with 1,411 expert annotations, 630 capability questions, and adversarial probes grounded in the misinformation effect, anchoring bias, and sycophancy, scored using an MQM framework. We further propose the Admittance-Resistance-Inductance (A-R-I) framework, which decomposes figure-reading trustworthiness into three behavioural axes scored through a dual-judge pipeline validated against human annotators (ICC = .91).

Evaluating thirteen frontier models, spanning open-weight architectures from 4B to 235B parameters alongside leading closed-source systems, we find that behavioural profiles diverge sharply from accuracy rankings. GPT-5.2 leads on description quality but scores only .56 on resistance to false premises, the Gemma family fabricates with near-zero admittance regardless of scale, numerical precision and visual counting account for the majority of errors, and description and comprehension prove to be partially dissociated. These findings demonstrate that trustworthy deployment requires behavioural profiling in addition to accuracy measurements.